

Demand-Side Data Collection: Sources, Provenance, and Manual Hand-offs

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1 Overview

This report documents the automated acquisition of the *demand-side* data inputs for Layer I (housing-demand estimation), covering the nine-county ABAG Bay Area (FIPS 06001, 06013, 06041, 06055, 06075, 06081, 06085, 06095, 06097). The data are *region-wide* (not per-locality) and therefore live at the Dropbox data root under `data/demand/`, resolved through the repository’s existing `paths.py/MFHR_DATA_ROOT` convention; the collection code lives in `demand_estimation/`. Of the twelve data sources, ten are **fully automated** here (ACS PUMS and aggregate tables, LEHD LODES8, TIGER/Line, USDA SSURGO, CGS seismic hazard, Gov-OPR statewide zoning, CPAD open space, CDE schools, GTFS transit, and IRS migration). Two are licensed and could not be automated: **CoreLogic** (the price/transaction spine) and **RS Means** (the construction-cost schedule that turns the soil instrument into a predicted-cost index). **CoreLogic has since been pulled manually and processed** (Section 5); **RS Means is now the single outstanding manual blocker** (Section 7). A third item, Infutor/Verisk migration, is an optional licensed upgrade for which a free IRS backup was pulled automatically.

Every fetch is idempotent, checksummed, and logged to `data/demand/_manifest.csv`. In total 1.5 GB was pulled across the automated sources.

2 Automated sources (inventory)

Table 1 is the source inventory: one row per automated source, with counts, sizes, and statuses read directly from the manifest. Status `downloaded/cached/verified` means data is present; `needs_key` means automatable but pending a credential; `partial` means some artifacts of a multi-part source landed. (USDA SSURGO appears as two rows — the tabular engineering extract and the spatial polygons that join it to block groups.)

Table 1: Automated demand-side source inventory (from `_manifest.csv`).

Source	Layer-I role	Vintage	Access	Endpoint	#	Size	Status	Chk
ACS PUMS	5-yr Household micro-data (BLP random-coefficient moments)	2020–2024 ACS 5-yr	FTP bulk (keyless)	<code>www2.census.gov/programs-surveys/acs/data/pums/2024/5-Year</code>	2	350.4 MB	cached	sha256
ACS aggregate tables	Tract/BG tenure, value, rent, income, structure	2023 ACS 5-yr	Census Data API (key)	<code>api.census.gov/data/2023/acs/acs5</code>	2	697.6 KB	cached	n/a
LEHD LODES8	Job access JA_{ℓ} (agglomeration coupling)	LODES8 v8.3 (2022, 2015)	direct download	<code>lehd.ces.census.gov/data/lodes/LODES8/ca</code>	7	135.3 MB	cached, downloaded	sha256
TIGER/Line geography	Geography spine; BG↔jurisdiction overlay	TIGER2023	direct download	<code>www2.census.gov/geo/tiger/TIGER2023</code>	5	176.0 MB	cached	sha256
USDA SSURGO (tabular)	Instrument: shrink-swell/plasticity/bearing (mukey)	SDA current (2025) ingverest	Soil Data Access REST	<code>sdmdataaccess.sc.egov.usda.gov/Tabular/post.rest</code>	12	2.1 MB	cached, downloaded	sha256
SSURGO (spatial)	Mapunit polygons — the mukey→BG join	SDA current	Soil Data Access REST	<code>sdmdataaccess.sc.egov.usda.gov/Tabular/post.rest</code>	11	196.6 MB	cached	sha256

continued...

(Table 1 continued)

Source	Layer-I role	Vintage	Access	Endpoint	#	Size	Status	Chk
CGS seismic hazard	Instrument (seismic capitalization)	control CGS cur-	ArcGIS Feature Service	services2 . com / Geohazards	2	22.3 MB	cached	sha256
Gov-OPR statewide zoning	By-right envelope backstop (Layer II)	2022-23 snapshot	ArcGIS Feature Service	services8 . com / California Statewide Zoning_North	1	453.1 MB	cached	sha256
CPAD open space	Amenity (proximity to protected areas)	control CPAD 2024a	ArcGIS Feature Service	services1 . com / cpad_2024a_unitsgdb	1	32.4 MB	cached	sha256
CDE schools	Amenity: school enrollment / quality proxy	CDE cur-	bulk flat file	www3.cde.ca.gov / demo-downloads / ce / cenroll12324 . txt	1	19.3 MB	cached	sha256
GTFS transit	Amenity: transit access (all operators via 511)	transit agency current	511 API + per-agency	api . 511 . org / transit / datafeeds	3	77.7 MB	cached, downloaded	sha256
IRS SOI migration	Moving-cost backup (free; Infutor/Verisk upgrade)	2021-2022	direct download	irs . gov / pub / irs-soi / countyinflow2122 . csv	2	8.5 MB	cached	sha256
Crime (SF+Oak+FBI)	Location amenity: incident BG (SF, Oakland) + FBI county	dis- FBI 2022	Socrata + FBI CDE + geocoder	data . sfgov . org ; api . usa . gov / crime/fbi	4	9.7 MB	cached, downloaded	sha256

3 Dataset granularity & key fields

Table 2 states, for every dataset, its *unit of observation* (the grain at which each row sits) and the key fields carried, so it is explicit what information each source contributes and at what level it must be aggregated to reach the block-group estimation geography.

Table 2: Granularity (unit of observation) and key fields by dataset.

Dataset	Grain (unit of obs.)	Key fields
ACS PUMS (housing)	household (weighted)	record TEN, HINCP, VALP, RNTP, BDSP, NP, BLD, YRBLT; PUMA, bay_area flag
ACS PUMS (person)	person (weighted)	record AGE, RAC1P, SCHL, HISP, SEX; PUMA
ACS tables	block group & tract	tenure / value / rent / income / structure counts (8 table groups), keyed by GEOID
LEHD LODES8	workplace block group	block → jobs by block (C000); distance-decay job access JA_ℓ
TIGER/Line	polygons: tract, BG, PUMA	county, place, GEOID, geometry; the BG↔jurisdiction crosswalk
SSURGO (tabular)	map unit (mukey) × soil horizon	lep_r (shrink-swell), pi_r, ll_r, K_{sat} , drainage, taxonomy

Dataset		Grain (unit of obs.)	Key fields
SSURGO	(spa- tial)	map-unit polygon	mukey , geometry ; areal weights to block groups
CGS hazard		hazard-zone polygon	seismic / liquefaction zones, Alquist–Priolo fault traces → BG area share
Gov-OPR zoning		zoning polygon (parcel-level)	County , Jurisdiction , zoning Code
CPAD	open	protected-area polygon	access type, managing agency → BG distance to nearest space
CDE schools		school × year	cumulative enrollment by school
GTFS transit		transit stop	stop_lat , stop_lon → BG stop density (1 km)
IRS migration		county-pair × year	inflow / outflow returns, exemptions, AGI
Crime		incident → BG (SF, Oakland); county (FBI, all 9)	SF/Oakland incidents geocoded to BG; FBI violent+property county rate per 100k
CoreLogic deed		transaction (recorded sale)	SALE_AMOUNT , SALE_DERIVED_DATE , deed/sale-type codes, resale/new-construction/interfamily/REO indicators
CoreLogic property	prop- erty	parcel (CLIP)	land use, parcel lat/long, assessed land/improvement/total value, tax, beds/baths/rooms, living & lot sq ft, units, stories, foundation, owner-occupancy

4 Per-source detail

4.1 ACS 5-year PUMS (household micro-data)

The household side of BLP. Downloaded via the *keyless* Census FTP bulk path (`csv_hca.zip` housing + `csv_pca.zip` person, California, 2020–2024 5-year), so no API key is required. The build identifies the 73 2020 PUMAs intersecting the nine counties (spatial join of the TIGER PUMA20 layer against the county layer) and tags each CA record with a `bay_area` flag, writing `ca_pums_household.parquet` (795,209 records; 193,449 Bay Area) and `ca_pums_person.parquet`. Variables retained: `TEN`, `HINCP`, `VALP`, `RNTP`, `BDSP`, `NP`, `BLD`, `YRBLT` (housing) and `AGEP`, `RAC1P`, `SCHL`, `HISP`, `SEX` (person), plus `PUMA`, `ST`, `SERIALNO` and the survey weights `WGTP`/`PWGTP`. All moments below are **WGTP-weighted**, hence population-correct (the weighted household total is 3.3 M, matching the nine-county count).

Table 3: ACS PUMS household summary (Bay Area; **WGTP**-weighted).

Variable	Households (wtd)	Median	Mean	P10	P90
Household income (HINCP)	3,275,379	\$115,000	\$166,064	\$22,200	\$360,000
Home value, owners (VALP)	1,877,800	\$900,000	\$1,170,785	\$350,000	\$2,100,000
Gross rent, renters (RNTP)	1,350,137	\$2,000	\$2,209	\$800	\$3,700
Bedrooms (BDSP)	3,324,794	3	2.7	1	4
Household size (NP)	3,275,379	2	2.6	1	5
Owner-occupied			57.3% of occupied		
Renter-occupied			42.7% of occupied		

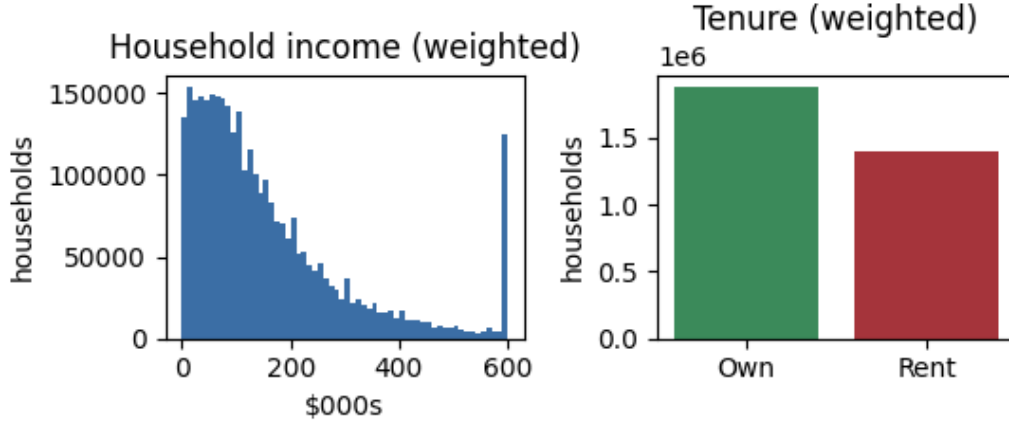


Figure 1: PUMS Bay Area households (weighted): income distribution (left) and tenure (right).

4.2 ACS aggregate tables

Tract- and block-group-level tenure, value, rent, income, and structure — the market shares and outside-option denominators of demand — pulled from the Census Data API (2023 ACS 5-year) for all nine counties: groups B25003 (tenure), B25077 (value), B25064 (rent), B19013 (income), B25024 (units in structure), B25118 (tenure by income), B25034 (year built), B25040 (heating fuel). The key is resolved from the environment / `CENSUS_API_KEY.txt`; if absent or rejected the step degrades to `needs_key` without blocking the run.

Table 4: ACS block-group aggregates (distribution across block groups).

Across block groups	N	Median	Mean	P10	P90
Owner-occupancy share (%)	5,163	63.2%	58.3%	16.6%	91.8%
Median home value	4,778	\$1,112,500	\$1,175,612	\$588,530	\$2,000,001
Median gross rent	4,260	\$2,600	\$2,592	\$1,679	\$3,501
Median household income	4,903	\$139,063	\$146,032	\$72,188	\$250,001

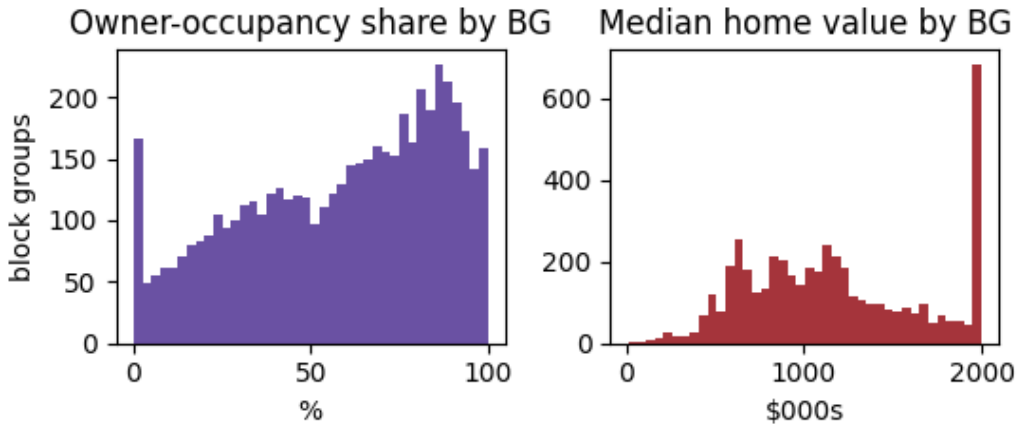


Figure 2: ACS block groups: owner-occupancy share (left) and median home value (right).

4.3 LEHD LODES8 (job access)

The agglomeration coupling: job access JA_ℓ is the empirical analogue of the toy model’s regional wage $w_m = w(N)$. WAC (workplace jobs by block), RAC, OD (main+aux), and the block crosswalk for California were downloaded for the panel year 2022 plus a 2015 pre-period for robustness. The build aggregates WAC jobs to block groups and computes a distance-decay job-access measure $JA_i = \sum_j W_j e^{-d_{ij}/d_0}$ ($d_0 = 10$ km, great-circle between BG centroids in CA-Albers), plus 10 km/30 km job counts, over the 5,187 Bay Area block groups ($\sum_j W_j = 4,015,993$ jobs).

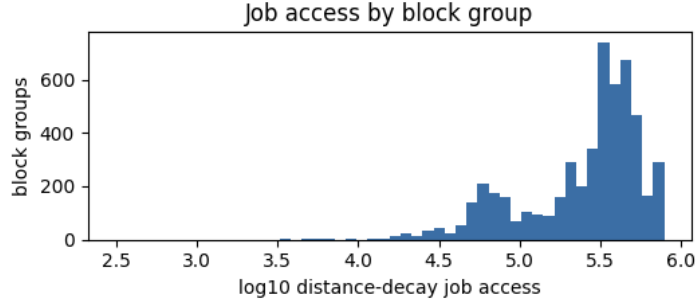


Figure 3: Distance-decay job access by block group (log scale).

Table 5 collects the block-group covariates contributed by this and the next few sources (LODES, SSURGO, CGS, CPAD, GTFS) in one place.

Table 5: Block-group covariate summary (LODES job access, SSURGO soil, CGS / CPAD / GTFS controls).

Block-group covariate	N	Median	Mean	P10	P90
Jobs located in BG (LODES)	5,187	193	774	44	1,363
Job access (decay-weighted)	5,187	326,589	320,037	60,884	545,295
Jobs within 30 km	5,187	1,415,136	1,186,822	203,833	1,815,310
Seismic-zone area share (CGS)	5,187	0	0.1	0	0.8
Open-space distance, m (CPAD)	5,187	285	382	31.3	749
Transit stops ≤ 1 km, all ops	5,187	20	40.0	0	93
Soil expansive share (SSURGO)	4,075	0	0.2	0	0.9
Soil linear extensibility, lep	4,075	3.6	3.5	0	7.3

4.4 TIGER/Line geography & the BG $\leftrightarrow$ jurisdiction crosswalk

Counties, tracts, block groups, places, and PUMAs (TIGER2023). The single most load-bearing derived artifact is `geography/crosswalks/bg_to_jurisdiction.parquet`: each of the 5,187 block groups is assigned to its dominant incorporated place by areal overlap (intersection in CA-Albers; dominant place = max intersection area), or to the “Unincorporated *County*” remainder when no place covers a majority of its area. This operationalizes the demand memo’s design in which jurisdiction is an *attribute of the location*; 188 distinct jurisdictions result.

Table 6: Geography by county (block groups, jurisdictions, unincorporated BGs).

County	Block groups	Jurisdictions	Uninc. BGs
Alameda	1,134	21	18
Contra Costa	708	47	51
Marin	181	23	22
Napa	108	8	29
San Francisco	681	1	0
San Mateo	500	30	37
Santa Clara	1,173	25	48
Solano	298	10	43
Sonoma	404	24	132
All 9 counties	5,187	188	380

4.5 USDA SSURGO (the instrument)

The cost-shifter instrument’s raw material. Rather than the multi-GB gSSURGO geodatabase, we use the USDA Soil Data Access (SDA) tabular REST service, which returns exactly the engineering fields needed without a spatial download. The nine counties are covered by 11 SSURGO survey areas (12 county overlaps; survey CA689 spans both San Francisco and San Mateo), discovered at run-time from the County-or-Parish overlap (symbols are not all FIPS-aligned — Alameda/SF/San Mateo/Santa Clara use legacy survey symbols). For each survey area we pull `mapunit` → `component` → `chorizon` linear extensibility (`lep_r`, shrink-swell / COLE proxy), plasticity index (`pi_r`), liquid limit (`ll_r`), K_{sat} , bulk density, drainage class, and taxonomy. The build aggregates horizons of the dominant component to a depth-weighted (0–150 cm) profile, yielding `soil/derived/mapunit_engineering_properties.parquet` (1,650 map units, with a derived shrink-swell class).

Map-unit → block-group join (now built). The tabular SDA route returns no geometry, so a second pull (`ssurgo_spatial.py`) fetches the SSURGO spatial MUPOLYGON mapunit polygons — 39,021 polygons across the same 11 survey areas — via SDA’s WKT export, cursor-paged on `mupolygonkey` (the public per-survey Web Soil Survey zips are not reliably addressable without a per-survey target date; SDA is the keyless equivalent). Polygon `mukeys` fully cover the tabular extract’s `mukeys`. The build areal-intersects the polygons with the block groups in CA-Albers and writes `soil/derived/bg_soil_engineering.parquet`: each of the 5,184 covered block groups carries the area-weighted `lep/pi/ll/ksat`/bulk-density, the expansive (high + very-high shrink-swell) area share, a coverage fraction, and an area-weighted missingness flag (mean coverage 99.7% of BG area). This resolves the BG-aggregation TODO the first run flagged; see Section 6 for the one remaining blocker.

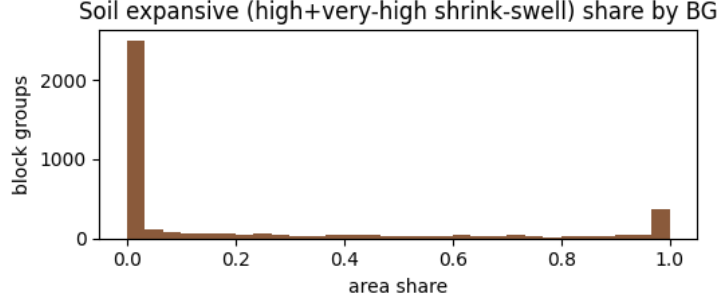


Figure 4: Soil expansive (high + very-high shrink-swell) area share by block group — the instrument’s raw spatial variation.

4.6 CGS seismic hazard (control)

Seismic risk is a *control*, not an instrument: it must be partialled out so the soil instrument is not contaminated by the seismic-capitalization channel. The statewide CGS Seismic Hazard Zones (zones of required investigation, incl. liquefaction and earthquake-induced landslides) feature layer is pulled clipped to the Bay Area bounding box, alongside the public Alquist–Priolo fault traces. The build computes each BG’s share of area inside a hazard zone.

4.7 Gov-OPR statewide zoning (Layer II backstop)

The by-right envelope $\bar{z}_j$ backstop — a single 2022–23 aggregated statewide snapshot from the Governor’s Office of Planning & Research, North half (which covers the Bay Area), clipped to the nine counties. This feeds Layer II, not Layer I; it was grabbed now because it is cheap while pulling spatial layers. *Caveat*: a snapshot, not a panel — flag for per-city fidelity checks.

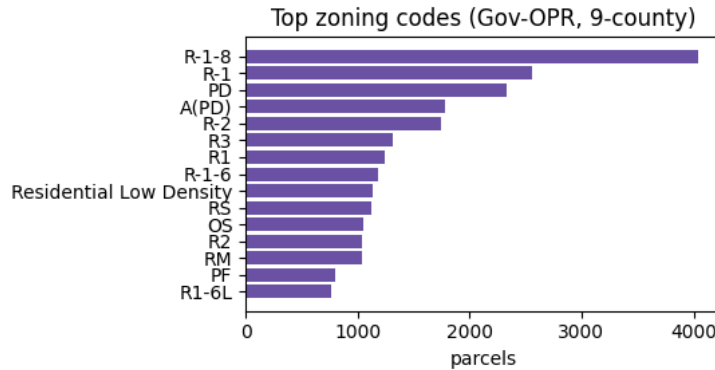


Figure 5: Most common local zoning codes (Gov-OPR snapshot, 9-county; 87,329 parcels, 4,725 distinct codes).

4.8 Amenity layers (controls)

Open space (CPAD 2024a) — like seismic, an amenity *control*, so the instrument can be residualized against proximity to protected areas; the build computes BG-centroid distance to the nearest protected area. **Schools (CDE)** — enrollment flat file (school-quality / demographics

proxy), discovered from the CDE data-download page. **Transit (GTFS).** The build counts GTFS stops within 1 km of each BG centroid. Two measures are kept for auditability: a keyless BART + Caltrain baseline (`transit_stops_1km_bartcaltrain`, 713 BGs with a stop — essentially rail stations) and, now that a 511 Bay Area API key is available, the full all-operator measure (`transit_stops_1km_all`). The latter is built from the 511 regional GTFS bundle (`operator_id=RG` — Muni, AC Transit, VTA, SamTrans, and every other operator; 21,028 stops), fetched into `amenities/transit/bay511/` and unioned with the per-agency feeds: it reaches 4,590 BGs, a roughly six-fold gain over the rail-only baseline. (Caltrain’s standalone `caltrain.com` feed now returns HTML, so the keyless baseline uses the Trillium Caltrain mirror; downloads are validated as real zips before use.)

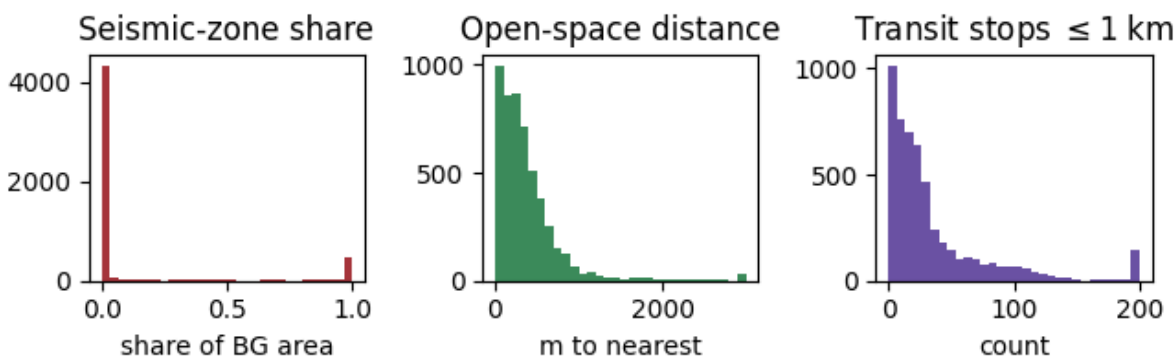


Figure 6: Instrument controls by block group: seismic-zone area share (CGS), distance to nearest protected open space (CPAD), and transit-stop density within 1 km (GTFS, all operators).

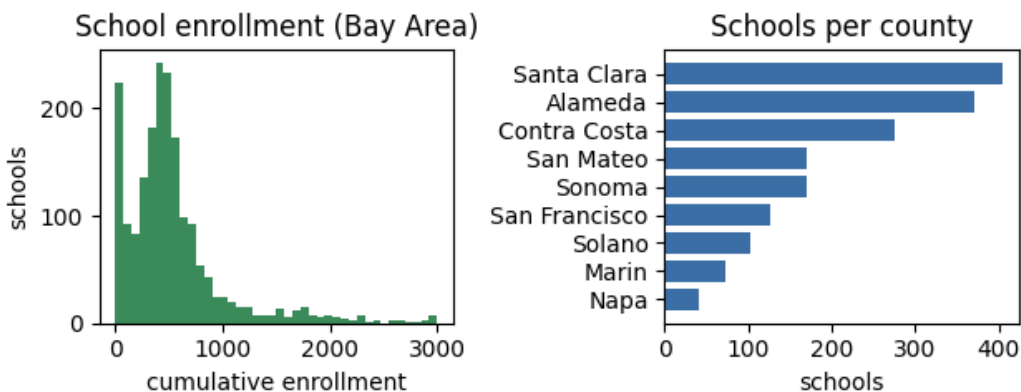


Figure 7: CDE schools: cumulative-enrollment distribution and school counts by county (1,720 Bay Area schools).

4.9 Crime (location disamenity)

Crime is a classic location (dis)amenity in housing demand — a willingness-to-pay shifter and a control. Coverage is assembled in two layers:

- **Incident-level, geocoded to block group:** San Francisco PD (DataSF / Socrata, with

coordinates) and Oakland PD (data.oaklandca.gov, addresses geocoded via the free Census batch geocoder) — 1,389,347 incidents 2019–2024 across 1,068 block groups, written to `crime/bg_crime.parquet` as `crime_per_1k_hh_yr`.

- **County-level, all nine counties (full coverage):** the FBI Crime Data Explorer (api.data.gov key) — violent + property offenses summed over every agency in each county yield a rate per 100k (Table 7), broadcast to every block group as `county_crime_rate_per_100k`.

Both layers are merged into the controls. San Jose’s only open feed is “calls for service” (police activity, not crime reports) behind a portal interstitial, so San Jose is covered by the FBI Santa Clara county rate rather than incident-level. The rankings are as expected — San Francisco and Alameda (Oakland) highest, Sonoma/Napa/Marin lowest.

Table 7: County crime rates (FBI CDE, 2022): violent + property offenses per 100k.

County (FBI 2022)	Violent	Property	Rate/100k
San Francisco	5,455	48,771	7,091
Alameda	11,022	68,638	4,918
Solano	2,506	12,047	3,229
Santa Clara	7,426	48,086	3,011
Contra Costa	4,789	24,528	2,530
San Mateo	2,291	15,685	2,514
Marin	646	4,449	1,968
Napa	453	2,049	1,854
Sonoma	1,581	5,791	1,523

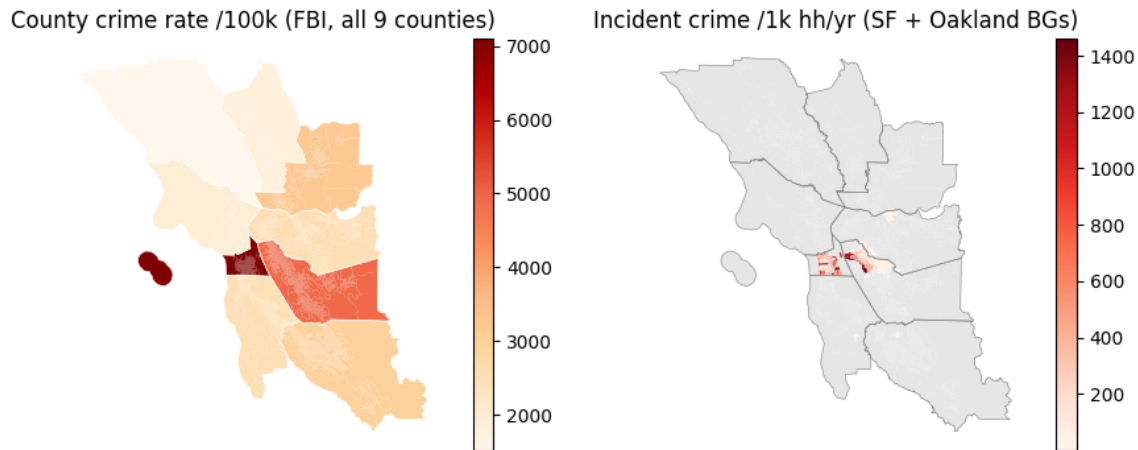


Figure 8: Crime by block group: county rate per 100k (FBI, all nine counties — full coverage, left) and incident crime per 1,000 households/year (SF + Oakland block groups, right).

4.10 IRS migration (moving-cost backup)

IRS SOI county-to-county inflow/outflow (2021–2022), the free automatable first-pass moving-cost input, filtered to the nine counties. The licensed Infutor/Verisk upgrade is stubbed (Section 7). Every Bay Area county except Solano shows net domestic out-migration in 2021–22 — the documented post-pandemic pattern.

Table 8: IRS county migration, 2021–22 (tax returns).

County	In-returns	Out-returns	Net
Alameda	40,374	51,822	-11,448
Contra Costa	24,234	28,095	-3,861
Marin	5,135	6,044	-909
Napa	2,293	2,576	-283
San Francisco	31,061	35,504	-4,443
San Mateo	20,241	24,779	-4,538
Santa Clara	41,725	53,787	-12,062
Solano	9,739	9,617	+122
Sonoma	7,954	7,993	-39
All 9 counties	182,756	220,217	-37,461

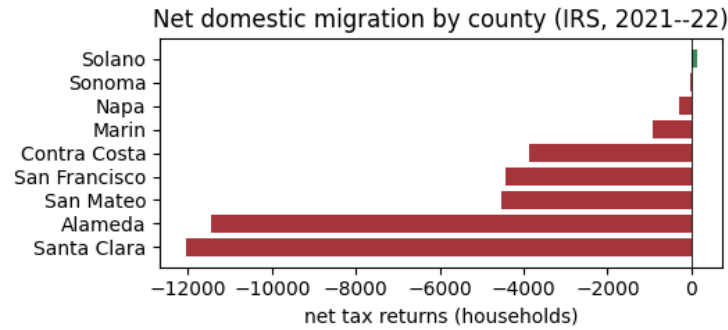


Figure 9: Net domestic migration by county (IRS SOI, 2021–22).

4.11 Spatial patterns

Figure 10 maps the key block-group variables. The expensive peninsula / San Francisco / Marin core, the San Francisco + Silicon Valley job-access peaks (the agglomeration centre), the low-ownership rental cores, and — importantly for identification — the soil instrument’s *independent* spatial variation (inland and east-bay expansive clays, not aligned with the price/income gradient) are all visible.

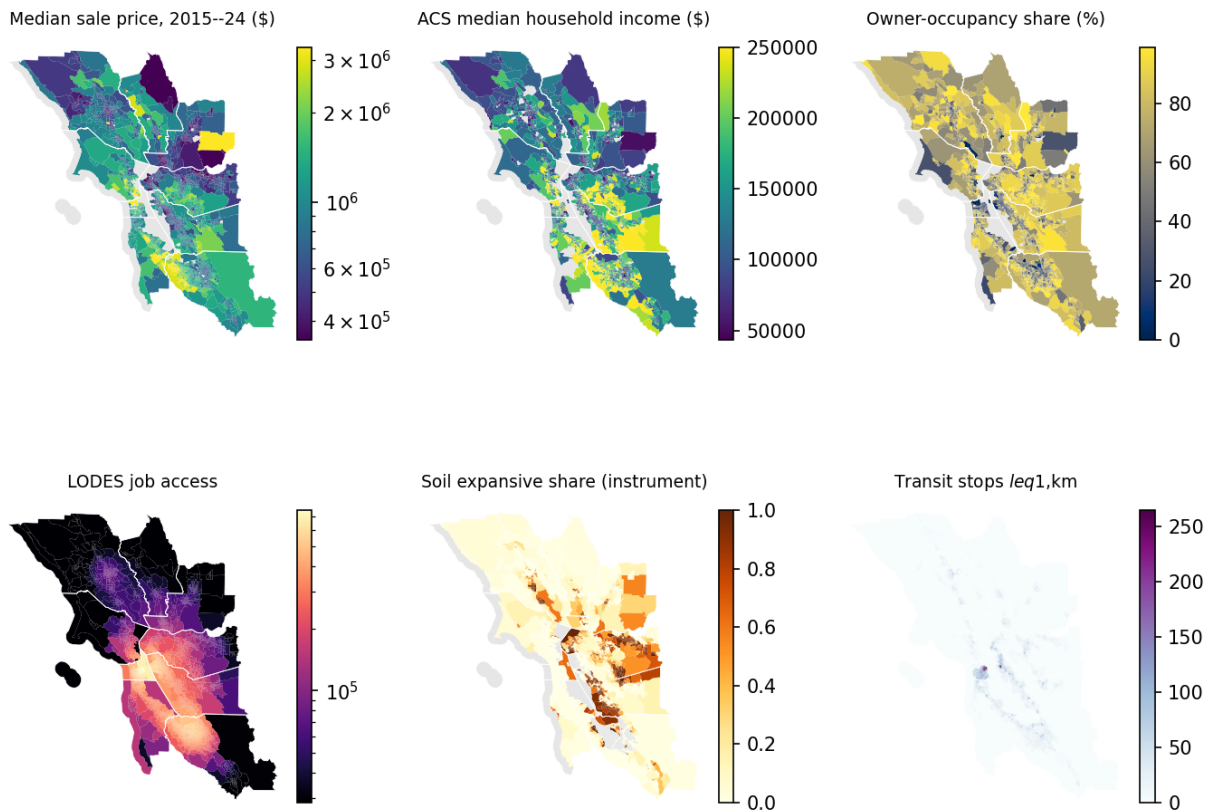


Figure 10: Block-group choropleths (county boundaries in white): median sale price (2015–24, CoreLogic), ACS median household income, owner-occupancy share, LODES job access, SSURGO soil expansive share (the instrument), and transit-stop density. Colour scales are 2nd–98th percentile; price and job access are log.

5 CoreLogic / Cotality property records (the price spine — landed)

The price/transaction spine has now landed. It was pulled manually from Stanford Libraries’ Redivis “Data Farm” (Cotality Smart Data Platform, formerly CoreLogic) as two Bay-Area-filtered extracts (the nine-county FIPS filter run server-side); PII columns (owner / buyer / seller names and mailing addresses) were excluded at query time per the Stanford EULA and IRB rules. The two tables sit at different grains:

- **Owner Transfer (deed)** — *transaction grain*: 5,337,067 recorded sales spanning 1900–2024, with sale amount/date, deed and sale-type codes, and the resale / new-construction / interfamily / foreclosure-REO indicators used to screen for arms-length transactions.
- **Property (tax assessor)** — *parcel grain*: 2,275,275 parcels (1,908,377 residential), each carrying land use, parcel lat/long, assessed land/improvement/total value, annual tax, beds/baths/rooms, living and lot square footage, units, stories, foundation type, and owner-occupancy.

Merge & geocoding. Each parcel is geocoded to a TIGER block group by its parcel lat/long (98.9% of parcels carry coordinates), and the deeds are joined to the parcels on CLIP: 100.0% of transactions match a parcel and 99.9% carry a block-group GEOID. The outputs are `corelogic/clean/corelogic_transactions_bg.parquet` (one row per sale, with characteristics + assessed value/tax + GEOID) and `corelogic/clean/corelogic_parcel_bg.parquet` (one row per parcel) — so every sale joins straight onto the crosswalk, job access, soil and controls.

First-pass diagnostics (is the data sane?). The headline analysis subset — residential, arms-length, sale year ≥ 1990 , price in \$10k–\$50M — is 3,169,205 transactions. Tables 9 and 10 and Figure 11 summarize it.

Table 9: Residential arms-length sales by county (1990–2024).

County	Sales	Median	P25	P75	Median \$/sqft
Alameda	638,761	\$416,000	\$233,000	\$700,000	\$279
Contra Costa	641,110	\$377,000	\$210,000	\$625,000	\$215
Marin	124,439	\$627,500	\$355,500	\$1,050,000	\$351
Napa	61,724	\$400,000	\$220,000	\$646,500	\$230
San Francisco	220,049	\$645,000	\$340,500	\$1,075,000	\$502
San Mateo	276,116	\$590,000	\$329,000	\$1,005,000	\$382
Santa Clara	696,180	\$520,000	\$292,000	\$852,500	\$326
Solano	255,577	\$285,000	\$165,000	\$450,000	\$169
Sonoma	255,249	\$350,155	\$205,000	\$575,000	\$225
All 9 counties	3,169,205	\$435,000	\$245,000	\$743,000	\$273

Table 10: Residential parcel characteristics (Property file).

Variable	N	Median	Mean	P10	P90
Year built	1,862,204	1,971	1,969	1,930	2,004
Bedrooms	1,800,324	3	3	2	4
Bathrooms	1,845,982	2	2	1	3
Living area (sq ft)	1,867,953	1,594	1,812	984	2,802
Lot size (sq ft)	1,751,679	6,098	46,199	2,500	20,473
Assessed total value	1,890,940	\$558,000	\$764,236	\$144,284	\$1,501,236
Annual property tax	1,888,307	\$7,581	\$9,995	\$2,535	\$18,991

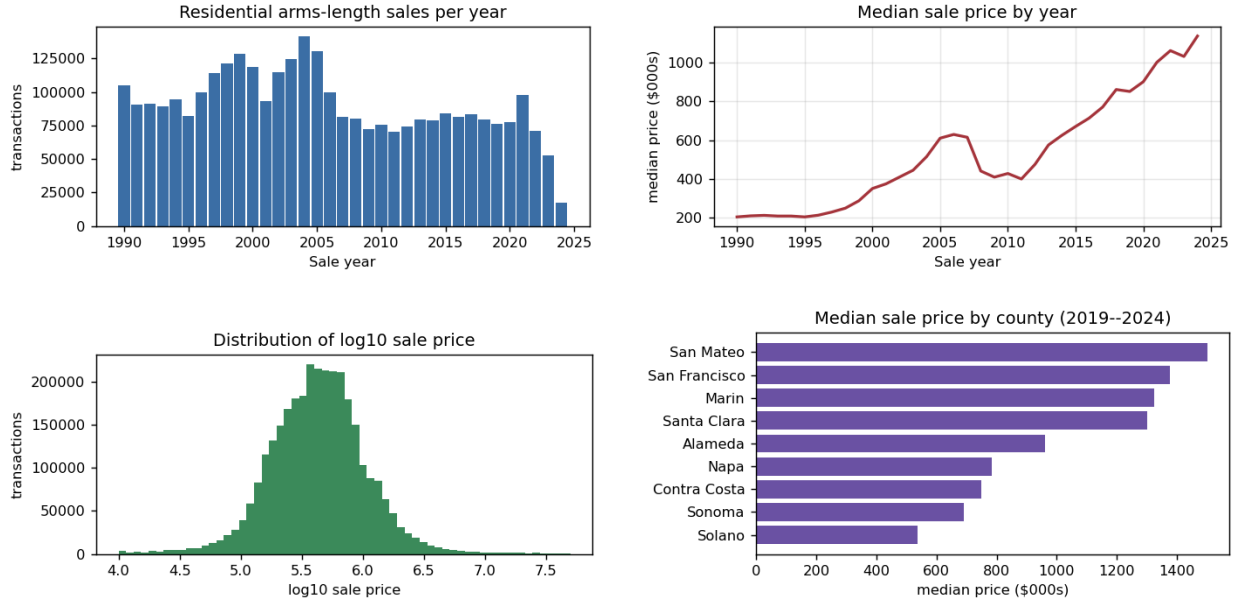


Figure 11: CoreLogic diagnostics (residential arms-length subset). Clockwise from top-left: sales per year, median sale price by year, median price by county (2019–2024), and the log-price distribution.

Verdict. The extracts are internally consistent and track external benchmarks. The median-price-by-year series (Figure 11, top-right) reproduces the Bay Area cycle — roughly \$200k in the early 1990s, a ~\$620k peak in 2006–07, the 2008–11 trough near \$410k, and the 2012–2022 boom past \$1M. The cross-county ordering (Marin / San Francisco / San Mateo / Santa Clara dear, Solano cheapest), the price-per-square-foot gradient, and the Prop-13-depressed assessed values (median assessed \$558k at a ~1.2% effective tax rate) are all what one expects. This is real signal, not noise — a sound basis to iterate the cleaning and arms-length definitions.

6 The instrument pipeline status

This is the section most relevant to the meeting. The soil instrument is built in four stages; this run completes the first two and stages the fourth, leaving a single manual blocker:

1. **SSURGO engineering-property extract — DONE.** `mapunit_engineering_properties.parquet` (1,650 map units) holds the depth-weighted shrink-swell / plasticity / bearing properties that are the instrument’s raw material.
2. **Map-unit → block-group aggregation — DONE (was the open TODO).** `bg_soil_engineering.parquet` areal-weights the engineering properties into the 5,184 covered block groups, using the SSURGO spatial MUPOLYGON polygons (39,021 polygons, full mukey coverage of the tabular extract) intersected with the TIGER block groups in CA-Albers. The soil instrument now reaches the estimation geography. This previously-flagged TODO is resolved.
3. **Predicted construction-cost index — BLOCKED on RS Means (the only remaining blocker).** Converting engineering properties into a dollar foundation-cost index requires the RS Means foundation-cost schedule (slab-on-grade vs. pier/pile vs. mat), a licensed product

(Section 7). Until `foundation_cost_schedule.csv` lands, the index cannot be computed; this is documented in `soil/derived/TODO_bg_aggregation_and_cost_index.md` (whose BG-aggregation half is now marked done).

4. **Residualization — STAGED, not estimated.** `clean/bg_design_matrix.parquet` assembles the block-group covariates the demand memo (§5) residualizes the predicted cost index against — job access, seismic-zone share, open-space distance, transit-stop density (shore distance and slope/elevation still to be added). The regression is left as a TODO because its dependent variable (the cost index) is blocked on RS Means.

7 Manual hand-offs (action required)

RECEIVED — CoreLogic / Cotality — price / transaction spine

Layer I role: the dependent-variable side of demand.

Status: landed and processed (Section 5). Pulled manually from **Stanford Libraries' Redivis “Data Farm”** (Cotality Smart Data Platform) as Bay-Area-filtered Owner-Transfer (deed) and Property (tax-assessor) extracts, PII excluded at query time. Merged on CLIP and geocoded to block groups (5,337,067 transactions, 2,275,275 parcels).

Remaining (minor): the owner-side *user-cost* series still needs an interest-rate series (e.g. FRED 30-yr mortgage) combined with these prices and the assessed-tax field — a small automatable follow-up, not a blocker.

ACTION REQUIRED — RS Means — construction-cost schedule

Layer I role: converts the SSURGO engineering extract into the predicted foundation-cost index — the actual instrument.

Why manual: paid subscription product; not scraped.

Daniel's steps: pull residential foundation-type cost differentials (slab-on-grade vs. pier/pile vs. mat) from RS Means (BU library; else Questrom).

Drop path: `data/demand/soil/derived/foundation_cost_schedule.csv` with columns `soil_or_foundation_class`, `usd_per_unit_premium`.

Note: published geotechnical cost rules-of-thumb can stand in for a first pass; default to RS Means for the real index.

ACTION REQUIRED — Infutor / Verisk migration — optional upgrade

Layer I role: structural moving-cost estimation à la Coven (only if moving costs are estimated).

Why manual: licensed address-history panels.

Already mitigated: the free IRS SOI county migration backup was pulled automatically into `data/demand/migration/irs/`. License Infutor/Verisk only to upgrade; drop under `data/demand/migration/{infutor,verisk}/`.

8 From assembled data to an estimated demand system

Everything above delivers the *inputs* to Layer I; it does **not** estimate demand. No discrete-choice model has been run — not even a plain conditional logit. This section is the roadmap from the assembled artifacts to an estimated demand system, and flags the data inputs still missing. (Estimation and all modeling choices are deliberately left to be done by hand.)

8.1 Ingredients: in place vs. outstanding

Ingredient	Supplied by	Status
Agents (household micro)	ca_pums_household / person . parquet (WGTP-weighted)	in place
Location attributes x_ℓ	bg_design_matrix + bg_soil_engineering + bg_controls + ACS BG tables	in place
Prices / rents	corelogic_transactions_bg . parquet; ACS gross rent	needs a BG quality-adjusted (hedonic) index
Market shares $s_{\ell t}$	ACS occupied-household counts (B25003)	needs construction once the product is defined
Supply-cost instrument	SSURGO soil (BG) + slope/topography	needs RS Means \$ index (blocked); see §6
Owner user cost	CoreLogic price + assessed tax	needs an interest-rate series (FRED 30-yr) — small
Choice set / products	—	a modeling decision , not data

8.2 Roadmap

1. **Define the product and market.** Unit of choice: block group ℓ (baseline), or $\ell \times$ structure type (SF/MF), or $\ell \times$ tenure. Market t : a pooled cross-section (e.g. 2019–2023) or a panel. Define the *outside option* (e.g. reside outside the region).
2. **Construct market shares $s_{\ell t}$** from the ACS occupied-household counts (B25003) consistent with the product definition; the denominator (market size) fixes the outside-option share.
3. **Build the price/rent index.** From CoreLogic, a BG-level *hedonic* sale-price index controlling for beds/baths/sq ft/year-built; for renters, ACS gross rent (B25064) or CoreLogic rentals; for owners, convert price to a *user cost* (price + property tax + an interest-rate series).
4. **Specify utility.** $\delta_\ell = x_\ell \beta - \alpha \text{price}_\ell + \xi_\ell$, with the agglomeration term $w(N)$ and the density disamenity $g(D_\ell)$ entered explicitly; random coefficients drawn over the PUMS demographic distribution.
5. **Estimate, baseline $\rightarrow$ structural.** (a) a plain conditional logit as a transparent first pass (note IIA); (b) BLP random coefficients via Berry inversion + PUMS micro-moments (income $\times$ price, income $\times$ job-access, family $\times$ bedrooms, tenure $\times \dots$).
6. **Instrument the price.** Price is correlated with unobserved location quality ξ_ℓ . Supply-cost shifters: the SSURGO soil foundation-cost instrument (once RS Means gives the dollar index) and slope/topography (Saiz tradition); plus BLP differentiation instruments (Gandhi–Houde). Exclusion: soil affects sorting *only* through construction cost, residualized against the seismic / open-space / slope channels already collected as controls.
7. **Micro-moments.** Match the joint distribution of household demographics $\times$ chosen-location attributes (PUMS agents) to pin the random coefficients.

8. **(For counterfactuals, not demand alone) the supply equation.** H_ℓ as a function of $\bar{z}_j$ and construction cost (RS Means + permits) closes the equilibrium the preemption counterfactuals need.

8.3 Remaining data inputs

- **RS Means** construction-cost schedule — the soil instrument’s dollar form; the one blocker (§7).
- **Interest-rate series** (e.g. FRED 30-yr mortgage) for the owner user cost — small, automatable.
- **(Layer II / III, separate)** permits for the supply equation; the multi-jurisdiction discretionary-review (τ_j) panel and the by-right envelope ($\bar{z}_j$) for the regulation game.

8.4 Tooling

The artifacts are already close to the (`product_data`, `agent_data`) shape `pyblp` (Conlon & Gortmaker) expects — one row per product×market with shares, prices, characteristics, and instruments; PUMS as weighted agents. A conditional-logit first pass needs only the product table.

9 Reproducibility

- **Geographic scope:** FIPS 06001, 06013, 06041, 06055, 06075, 06081, 06085, 06095, 06097; national/statewide sources are filtered to these.
- **Manifest:** `data/demand/_manifest.csv` — columns `source`, `url`, `local_path`, `bytes`, `sha256`, `fetch_at`, `status`; rows keyed by (source, path) so re-runs update in place.
- **Paths:** `demand_estimation/demand_paths.py` imports `DATA_ROOT` from the minutes pipeline’s `paths.py`; respects `MFHR_DATA_ROOT`. No absolute paths are hard-coded.
- **Dependencies added:** `pyarrow`, `pyogrio` (bundles GDAL), `shapely`, `us` (see `requirements.txt` / `environment-notes.md`).
- **One command:** `python -m demand_estimation.build` (idempotent, resumable, failure-isolated per source).

10 Definition-of-done checklist

- ✓ Every §2 source downloaded, checksummed (`sha256`), and logged in `_manifest.csv`.
- ✓ BG↔jurisdiction crosswalk built (5,187 block groups → 188 jurisdictions) — the load-bearing artifact.
- ✓ Soil engineering-property extract built (1,650 map units), **and** the map-unit→block-group areal-weighted aggregate (`bg_soil_engineering.parquet`, 5,184 BGs) from 39,021 SSURGO spatial polygons (full mukey coverage). Only the RS Means cost-index join remains, as a documented TODO.
- ✓ Hazard + amenity controls staged for residualization (`bg_controls.parquet`: seismic-zone share, open-space distance, transit-stop density).
- ✓ ACS micro (795,209 CA households) + BG tables (8 table groups) staged for BLP.

- ✓ IRS migration pulled; Infutor/Verisk stubbed.
 - ✓ Every §3 manual source has a `_stubs/<name>/README.md` + manifest row (CoreLogic, RS Means, Infutor/Verisk).
 - ✓ CoreLogic/Cotality price spine **landed & processed** (5,337,067 sales, 2,275,275 parcels, merged + geocoded to BG, sanity-checked); RS Means remains the one outstanding licensed blocker.
 - ✓ 511 regional transit: key **present** $\Rightarrow$ all-operator bundle (21,028 stops) fetched; full measure `transit_stops_1km_all` (4,590 BGs) built, with the BART+Caltrain baseline (`transit_stops_1km_bartcaltrain`) retained for auditability.
 - ✓ `demand_data_report.tex` compiled to PDF, zero undefined references.
 - ✓ Nothing behind auth was scraped or circumvented; no credentials fabricated (CGS regulatory layer pulled from its public copy; 511 transit fetched with the supplied key; CoreLogic/RS Means stubbed, not sourced).
- *Single remaining soil blocker:* only the predicted construction-cost index (needs the RS Means schedule) is outstanding; the map-unit $\rightarrow$ block-group aggregation TODO from the first run is now resolved — see Section 6.